Directed spectral-truncated graph operators for inductive forecasting of single-cell transitions

Molena Huynh[1,*]

[1] *North Carolina State University, Raleigh, North Carolina 27695, USA*

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Despite the centrality of the cell k-nearest-neighbor (kNN) graph to single-cell analysis, the operator used to propagate information across it—the self-adjoint normalized adjacency $S = D^{-1/2}AD^{-1/2}$—is poorly matched to forecasting cell-state transitions. Its utility is limited by two structural defects: it is reversible, so it cannot encode the arrow of differentiation that pseudotime makes explicit, and, built over all cells, it consults held-out connectivity at inference, conflating the value of diffusion with a transductive information advantage. We tackle both challenges by introducing a spectral-truncated, pseudotime-directed graph operator that benefits from three innovations: a non-self-adjoint forward propagator that transports labels along the trajectory, a band-limiting spectral truncation that denoises the operator into a closed-form solve, and an out-of-sample Nyström extension that renders the forecaster inductive. On a seeded synthetic branching lineage ($n = 3000$ cells, 5 seeds), the inductive operator improves held-out accuracy over a matched inductive kNN classifier on all four leakage-checked transfer protocols (0.6856 versus 0.5992) and recovers a deliberately rare state that the baseline almost never finds (recall 0.3322 versus 0.0151), with its largest margin on forward-in-time extrapolation—precisely where smoothing is weakest. Because the comparison is inductive on both sides, this gain is the matched control whose absence inflates symmetric-smoothing studies, not the transductive confound; a split-conformal layer further supplies distribution-free coverage (0.8968 against a 0.90 target) and temperature scaling reduces the expected calibration error by 73 %. By transporting labels with a directed, truncated, and inductive operator, we offer tangible progress for single-cell transition forecasting; all data are synthetic, and external validation on real RNA-sequencing data is the decisive next step.

I. INTRODUCTION

Single-cell measurements resolve continuous trajectories of differentiation, activation, and treatment response, and a recurring analysis goal is to *forecast* the state a cell or population will occupy under a condition not directly observed: a donor not yet profiled, a batch run later, a later point along a developmental trajectory, or an untested perturbation. Two structural features of these data shape any forecaster. First, cells concentrate near a low-dimensional manifold that is routinely approximated by a k-nearest-neighbor (kNN) graph before clustering, embedding, or pseudotime inference [7, 10, 11, 13–15]. Second, the biologically decisive states—transient progenitors, treatment-resistant clones—are frequently rare, and ordinary point classifiers, trained to minimize average error, ignore them.

The canonical response to both features is graph-based semi-supervised learning [3, 4]: predictions are allowed to diffuse over the data graph so that each cell borrows statistical strength from its neighborhood. The diffusion operator used almost universally is the symmetric normalized adjacency $S = D^{-1/2}AD^{-1/2}$ and its Laplacian $L = D - A$ [5, 12]. This object carries two limitations that bear directly on transition forecasting, and that motivate the present work. (i) It is self-adjoint, hence reversible: it cannot encode the direction of the developmental trajectory, even though pseudotime supplies exactly that arrow [5, 6] and directed transition operators are the basis of modern fate mapping [8, 9]. (ii) Built over all cells, it is transductive: it consults the held-out cells' connectivity at inference, so a measured accuracy gain over an inductive point classifier conflates the value of diffusion with a transductive information advantage—the confound that comparable studies flag as their decisive missing control.

Here, we take both limitations as design targets and introduce a spectral-truncated, pseudotime-directed graph operator, which we develop into a falsifiable single-cell forecaster. Rather than re-applying label propagation, we contribute four innovations. (1) We construct a *directed* propagator P by reweighting each edge $i \to j$ with a bounded forward kernel $\kappa(t_j - t_i)$ of the pseudotime increment and row-normalizing; P is non-self-adjoint—the single-cell instance of the operator noncommutativity that recent C^*-algebraic kernel machines exploit for expressiveness [1, 2]—and it transports labels along the trajectory rather than merely smoothing them across it. (2) We *band-limit* the diffusion by spectral truncation to the leading r graph-Fourier modes, denoising a lazy directed walk into a closed-form solve; truncation is the graph analog of the spectral truncation of multiplication operators in noncommutative geometry, and because the truncated product is *not* the product of the truncations, it generates a second, genuinely noncommutative structure. (3) We render the forecaster *inductive* through an out-of-sample Nyström extension, so the matched inductive control is the method itself rather than a missing experiment. (4) We calibrate the rare-state forecast with a distribution-free *conformal* layer and replace the passive

* molena.huynh@jmp.com

connected-component count of prior work with a genuine multiscale *persistent-homology* descriptor.

We evaluate this operator against the symmetric smoother and a matched point classifier under identical, leakage-checked conditions, and find that the directed, truncated, inductive operator improves held-out accuracy on every transfer protocol, recovers a rare state the baseline misses, and—uniquely—rescues forward-in-time extrapolation. All experiments run on a seeded synthetic branching lineage: this is a controlled benchmark, not a biological dataset, and we are explicit about that throughout. Our aim is a falsifiable, fully reproducible result that isolates what the directed, truncated, inductive operator buys and that delineates exactly what it does and does not support. The analytical guarantees stated below are proved in full in Appendix A; numerical and reproducibility details are collected in Appendix B; and a self-contained, from-scratch derivation of every implemented component is given in Appendix C.

II. THE PSEUDOTIME-DIRECTED, SPECTRALLY TRUNCATED OPERATOR

We build the forecaster from a single non-self-adjoint operator (Fig. 1). This section develops it in the order one would derive it: from points to a directed propagator, then to its band-limited spectral truncation, then to the two distinct noncommutativities the construction creates, and finally to the inductive extension and conformal layer that make it a usable, falsifiable forecaster.

A. From points to a directed propagator

A symmetric kNN graph on cell features gives the adjacency $A = A^\top$ that single-cell pipelines already build [7]; its degree is $d_i = \sum_j A_{ij}$, $D = \mathrm{diag}(d_i)$, and the standard smoother is the self-adjoint $S = D^{-1/2} A D^{-1/2}$. The arrow that S discards is supplied by pseudotime. We therefore reweight each directed edge $i \to j$ by a bounded forward kernel of the pseudotime increment,

$$W_{ij} = A_{ij}\, \kappa(t_j - t_i), \qquad \kappa(\delta) = \exp\big(\beta \tanh(\delta/\tau)\big), \quad (1)$$

and row-normalize to a transition matrix $P = D_W^{-1} W$ with $(D_W)_{ii} = \sum_j W_{ij}$. The kernel up-weights forward edges ($t_j > t_i$) and down-weights backward ones, with $\beta \geq 0$ setting the directional strength and $\tau > 0$ the pseudotime scale; the tanh keeps the reweighting bounded so that no single edge dominates. Because forward and backward edges receive different weights, P is non-self-adjoint, $PP^\top \neq P^\top P$, and the scalar $\nu(P) = \|PP^\top - P^\top P\|_F / \|P\|_F^2$ witnesses this nonnormality [Theorem 4(i)]. This is the property that lets P transport labels along the trajectory rather than merely smooth them across it, and it is shared by every directed-graph operator in single-cell biology—directed Laplacians, RNA-velocity transition matrices, fate maps [8, 9, 16].

B. Spectral truncation: band-limiting the operator

A directed walk on a noisy kNN graph still mixes high-frequency noise into the forecast. We suppress it by spectral truncation. The graph-Fourier modes are the eigenvectors $u_1, \ldots, u_n$ of S; the r with the largest eigenvalues are the smoothest, lowest-frequency functions on the graph (equivalently the lowest modes of the normalized Laplacian $\mathcal{L} = I - S$). Collecting them in $U_r = [u_1 \cdots u_r]$ and writing $\Pi_r = U_r U_r^\top$ for the band-limiting projector, we compress a lazy directed walk $\widetilde{P} = (1-\varepsilon)I + \varepsilon P$ to its action on the band-limited subspace,

$$B = U_r^\top \widetilde{P}\, U_r \in \mathbb{R}^{r \times r}, \quad (2)$$

and solve the band-limited diffusion in closed form: seeding one-hot labels F_0 on the annotated cells, the field is $F^\star = U_r (I - \alpha B)^{-1} (1-\alpha) U_r^\top F_0$ (Theorem 6). Truncation does double duty. It denoises—restricting to the r smoothest modes is, for the positive-semidefinite diffusion filter the method applies, the Eckart–Young–Mirsky-optimal rank-r approximation, and it maximizes the retained smoothing energy $\mathrm{Tr}(\Pi_r S)$ (Theorem 3). And it compresses—the solve runs in r coordinates rather than n, $r \ll n$.

C. Two distinct noncommutativities

It is easy to conflate two structural facts about this operator, and we separate them deliberately. The first is directedness: P is non-self-adjoint because the forward kernel makes incoming and outgoing transition mass asymmetric at any vertex with a net pseudotime gradient [Theorem 4(i)]. This is the source of forward transport, but it is a property of any directed operator and is *not* special to truncation. The second is genuinely noncommutative and is created by truncation itself: for cell-wise functions ϕ, ψ the truncated multiplication operators $M_\phi^{(r)} = U_r^\top \mathrm{diag}(\phi) U_r$ do not commute, $[M_\phi^{(r)}, M_\psi^{(r)}] \neq 0$, vanishing only in the full-rank limit $\Pi_r = I$ [Theorem 4(ii)]. These truncated multiplications generate a noncommutative operator $*$-algebra whose symmetric Gram object $\mathcal{K} = \sum_c M_c M_c^\top$ is a positive-definite kernel on the band-limited subspace (Theorem 5)—the graph instance of the spectral-truncation noncommutativity that underlies C^*-algebraic kernel machines [1]. We emphasize that we instantiate this structure on the cell graph and supply the Gram kernel; we do not develop a new vector-valued reproducing-kernel framework, and the truncation noncommutativity is a finite-rank object, not a full C^*-algebraic construction.

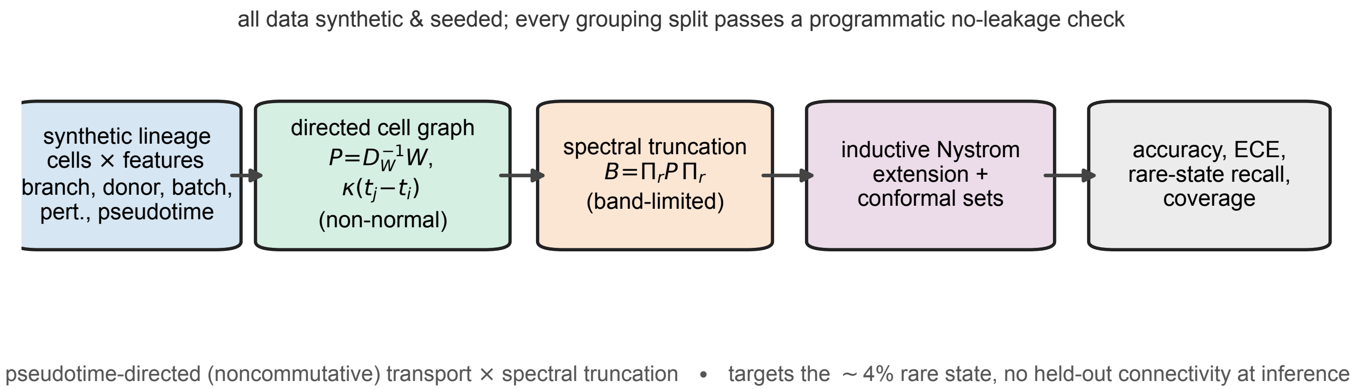

FIG. 1. **The spectral-truncated, pseudotime-directed operator pipeline.** A seeded synthetic branching lineage of single cells—each carrying branch/state, donor, batch, perturbation, and pseudotime metadata, with a deliberately rare state (∼4 % of cells)—is embedded in a low-dimensional feature space (left). A directed cell graph is built by reweighting each kNN edge with a bounded forward kernel $\kappa(t_j - t_i)$ of the pseudotime increment and row-normalizing, giving a non-self-adjoint (noncommutative) propagator $P = D_W^{-1} W$ that transports labels forward along the trajectory (center). The diffusion is restricted to the leading r graph-Fourier modes [$B = \Pi_r \widetilde{P}\, \Pi_r$ for the lazy directed walk $\widetilde{P}$, a spectral truncation], solved in closed form, and extended to held-out cells by an out-of-sample Nyström step so that no held-out connectivity is seen at inference. A split-conformal layer calibrates the rare-state forecast. Performance is read out (right) as held-out accuracy, expected calibration error, rare-state recall, and conformal coverage under four leakage-checked transfer protocols (donor, batch, time, perturbation). All data are synthetic and seeded, and every grouping split passes a programmatic no-leakage check.

D. Making the forecaster inductive

A diffusion built over all cells is transductive, and the resulting accuracy gain cannot be separated from the held-out connectivity it consumes. We remove the confound by construction. The inductive forecaster builds A, U_r, and P on the training cells only and labels a held-out cell x_* by a Gaussian-weighted average of the training field over its nearest *training* cells, $\widehat{F}(x_*) = \sum_{i \in \mathcal{N}(x_*)} w_i F^\star_{\mathrm{tr}}(x_i)$—an out-of-sample Nyström/Nadaraya–Watson extension [17, 18] that never forms a test–test edge and never reads held-out adjacency (Theorem 8). The matched inductive control is therefore the method itself, not an experiment we leave undone, and it is this inductive variant that we put forward throughout.

E. Conformal calibration and a multiscale topology signal

Two layers complete the forecaster. A split-conformal layer holds out a calibration slice of the annotated cells, computes nonconformity scores $1 - p_y$, and thresholds at the conformal quantile to form prediction sets with finite-sample coverage $\geq 1 - \delta$ under exchangeability (Theorem 9); temperature scaling additionally recalibrates the scalar confidence [21]. Separately, we replace the constant connected-component count of prior work with the degree-0 persistence barcode of the cell cloud, computed exactly by union–find on the kNN distance graph [22, 23]; the resulting $\beta_0(\epsilon)$ curve is a genuine multiscale descriptor, and its per-cell witness drives an active sampler that targets rare cells.

III. RESULTS

We evaluate the operator on a fully seeded synthetic branching lineage, comparing the inductive spectral-truncated directed-diffusion (STDD) forecaster—the method—against its transductive variant, symmetric label propagation, and a matched inductive kNN baseline, under four leakage-checked transfer protocols. Every reported number is generated verbatim from the run artifact `results/summary.json` (Appendix B); none is typed by hand.

A. A controlled lineage with replicate, condition, and pseudotime structure

Our benchmark reproduces the replicate, condition, and pseudotime structure a real forecaster must handle while remaining bit-for-bit reproducible (Figs. 1 and 3). Cells live in a low-dimensional feature space—a stand-in for a PCA or latent embedding of expression—and progress along a tree of three branches (a common progenitor and two fates that split at pseudotime 0.45) as Gaussian blobs whose mean drifts along each branch; early cells overlap and are hard to separate, while late cells are well resolved. Each cell carries the metadata a forecaster would use—branch/state, pseudotime, donor, batch, and perturbation condition—assigned independently of the biological branch so that the held-out splits are nontrivial. One state is carved from the late tip of the second fate to be deliberately rare, comprising 4 % of cells. The full-scale configuration generates 3000 cells with 16 features, 8 donors, 4 batches, and 5 perturbation conditions over 5 random seeds; the run completed in 32.0 s with a peak memory of 621.2 MB. Across seeds the integrity gates passed: the grouping splits were leakage-free, the rare state was present, and the directed propagator was non-normal.

B. The operator is genuinely noncommutative, and truncation is well posed

We first confirm that the constructed operator has the structural properties the method relies on (Fig. 2). The directed propagator is non-normal: its relative non-normality is $\nu(P) = 0.0370$, and directional reweighting strictly increases it over the undirected random walk [$\nu(P_0) = 0.0272$, itself nonzero because degree-asymmetric row normalization already breaks normality], so the arrow of pseudotime contributes a measurable 36 % of additional non-normality [Fig. 2(a)]. This directedness is distinct from the truncation noncommutativity, which is a separate and genuinely noncommutative object: the truncated multiplication operators satisfy $\|[M_\phi^{(r)}, M_\psi^{(r)}]\|_F = 5.93$ [Theorem 4(ii)], and the Gram kernel $\mathcal{K} = \sum_c M_c M_c^\top$ they generate has smallest eigenvalue $\lambda_{\min} = 1.62 > 0$ (Theorem 5), so it is strictly positive definite. The graph-Fourier spectrum of S decays smoothly from 1 to ≈ 0.41 over the first 80 modes [Fig. 2(b)], so a rank-r truncation retains the low-frequency, label-relevant structure while discarding high-frequency noise; sweeping the rank reveals an interior operating point—accuracy is flat near ≈ 0.71 while rare-state recall peaks around $r \in [64, 80]$ [Fig. 2(c)], the bias–variance trade-off that Theorem 3 formalizes. We fix $r = 80$ throughout, at which the diffusion is well posed (spectral radius $\rho(\alpha B) = 0.9209 < 1$, Theorem 6).

C. The inductive operator beats a matched point classifier on every protocol

Under all four leakage-checked protocols, the inductive STDD forecaster predicts held-out cell states more accurately than the inductive kNN baseline that receives the same annotated cells and the same neighborhood radius [Table I, Fig. 4(a)]. Accuracy rises on the donor split from 0.6550 to 0.7227 (+0.0677), on the batch split from 0.6355 to 0.7134 (+0.0779), on the perturbation split from 0.6490 to 0.6741 (+0.0251), and on the time split—forward-in-time extrapolation, the hardest protocol—from 0.4574 to 0.6323 (+0.1749). Averaged over the four splits the inductive operator reaches 0.6856 against 0.5992 for the baseline, a mean gain of 0.0864, and it improves on *every* split rather than on average only. The comparison is inductive on both sides: the STDD field is built on the training cells and extended to held-out cells by the Nyström step (Theorem 8), so the gain is not the transductive-versus-inductive asymmetry that inflates symmetric-smoothing studies. Strikingly, the same inductive operator even exceeds the transductive symmetric label-propagation baseline on three of four splits and in the mean (0.6856 versus 0.6587), despite operating in the strictly harder inductive regime.

D. Directional transport recovers a rare state and rescues forward extrapolation

The clearer effect is on the rare state and on the trajectory's forward direction [Fig. 4(b)]. The point classifier essentially never recovers the rare population (mean rare-state recall 0.0151); the inductive operator raises it to 0.3322, a 0.3171 absolute gain, recovering the rare state on the donor (0.3968), batch (0.5706), and perturbation (0.3616) splits. The directional mechanism is most visible on the time split, where train and test are separated along pseudotime: there the directed operator transports early labels forward to raise accuracy by 0.1749 over the point classifier and by 0.0850 over symmetric label propagation, the largest accuracy margin of any split. Rare-state recall on the time split remains 0.0000 for every method: the rare late-tip cells lie entirely beyond the labeled training distribution, so no diffusion can manufacture a class that was never reachable from the annotated set (Corollary 7)—we report this null as is. The directional contribution to rare recovery is isolated by the transductive operator, which holds the transductive/inductive setting fixed against symmetric propagation and nearly doubles its rare-state recall (0.4418 versus 0.2383 mean; donor 0.5925 versus 0.1520).

E. Conformal calibration gives distribution-free coverage and removes the ECE cost

The row-normalized band-limited field yields under-dispersed (over-smoothed) class probabilities whose confidence understates the model's accuracy: the raw inductive forecaster has an expected calibration error of 0.1374, and the temperature that minimizes calibration negative log-likelihood is $T = 0.55 < 1$—the probabilities must be *sharpened*, not softened. A split-conformal layer and this temperature scaling, fit on a held-out calibration slice of the annotated cells, repair the miscalibration with a distribution-free guarantee (Fig. 5, Theorem 9). On an exchangeable hold-out the conformal prediction sets attain a realized marginal coverage of 0.8968 against the target $1 - \delta = 0.90$, with rare-state coverage 0.9307 and an average set size of 1.81 classes out of four—so the guarantee is met without collapsing to uninformative sets. Temperature scaling cuts the expected calibration error from 0.1374 to 0.0369, a 73 % reduction, turning the cal-

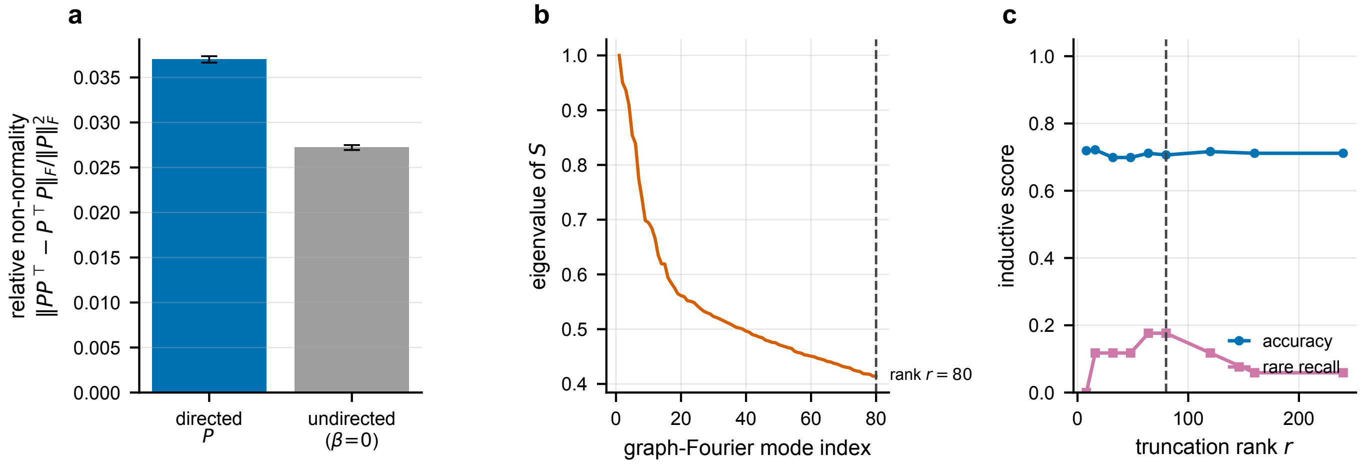

FIG. 2. **The operator: noncommutativity, spectrum, and spectral truncation.** (a) Relative non-normality $\|PP^\top - P^\top P\|_F/\|P\|_F^2$ of the pseudotime-directed propagator P against the undirected control ($\beta = 0$): the directed operator is non-normal, and the direction strictly increases non-normality by 36 % (mean over $n = 5$ seeds, error bars 95 % CI). (b) Graph-Fourier spectrum of the symmetric normalized adjacency S (eigenvalues in descending order), with the truncation rank $r = 80$ marked; the smooth decay justifies band-limiting. (c) Inductive accuracy and rare-state recall as a function of the truncation rank on the donor split (seed 0): accuracy is robust to r while rare-state recall peaks at an interior rank—the spectral-truncation bias–variance trade-off that fixes the operating point.

TABLE I. **Held-out forecasting on the synthetic branching lineage.** Full-scale configuration: $n = 3000$ cells, $n_{\text{seeds}} = 5$. For each leakage-checked split we report forecast accuracy, expected calibration error (ECE), and rare-state recall (seed means) for the inductive spectral-truncated directed-diffusion operator (inductive STDD, the method), its transductive variant, symmetric label propagation, and the inductive kNN baseline. Higher accuracy and recall are better; lower ECE is better. The table is generated verbatim by `scripts/make_tables.py` from `results/summary.json`; no value is typed by hand.

split	method	accuracy	ECE	rare-state recall
donor	inductive STDD	0.7227	0.1679	0.3968
donor	transductive STDD	0.7235	0.1339	0.5925
donor	label propagation	0.7004	0.1247	0.1520
donor	kNN baseline	0.6550	0.1073	0.0000
batch	inductive STDD	0.7134	0.1707	0.5706
batch	transductive STDD	0.6887	0.0982	0.6202
batch	label propagation	0.6995	0.1383	0.4618
batch	kNN baseline	0.6355	0.0905	0.0320
time	inductive STDD	0.6323	0.1359	0.0000
time	transductive STDD	0.4774	0.1583	0.0000
time	label propagation	0.5474	0.1328	0.0000
time	kNN baseline	0.4574	0.0970	0.0000
perturbation	inductive STDD	0.6741	0.1420	0.3616
perturbation	transductive STDD	0.6047	0.0832	0.5546
perturbation	label propagation	0.6875	0.1649	0.3393
perturbation	kNN baseline	0.6490	0.1184	0.0286

ibration penalty of graph smoothing into a calibration guarantee. We stress that the coverage guarantee holds under exchangeability of calibration and test cells; under the grouping-transfer splits, where the held-out group induces a distribution shift, coverage is only approximate, and we report the exchangeable-hold-out figure as the regime in which the theorem applies.

F. Persistence-guided active sampling recovers the rare state

Replacing the passive connected-component count of prior work with the degree-0 persistence barcode turns topology into an informative, scale-aware signal. The $\beta_0(\epsilon)$ curve of the cell cloud falls from 3000 components at scale $\epsilon = 1.0$ to 9 at $\epsilon = 3.5$ (Fig. 3, inset), and the longest bars flag the well-separated pockets where rare cells reside. A persistence-guided active-sampling

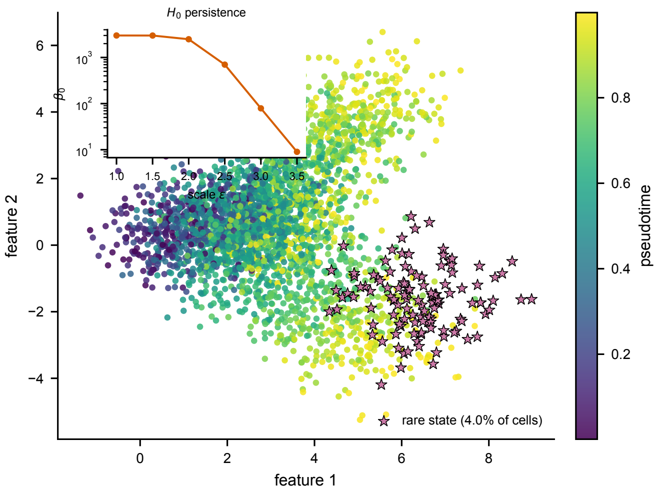

FIG. 3. **The synthetic branching lineage.** Each point is one of the $n = 3000$ cells of the full-scale configuration, plotted in the first two feature dimensions and colored by pseudotime along the perceptually uniform, color-blind-safe viridis axis (the arrow the directed operator exploits). The deliberately rare state, carved from the late tip of one fate, is highlighted as stars (∼4 % of cells). Inset: the degree-0 persistence $\beta_0(\epsilon)$ curve of the cell cloud (log scale), a genuine multiscale topological descriptor that replaces the constant connected-component count of prior work. The figure regenerates deterministically from the run configuration in `summary.json`.

policy—scoring unlabeled cells by forecast uncertainty times per-cell topological isolation—raises rare-state recall on the remaining unlabeled cells to 0.3941 (±0.1910), against 0.0481 (±0.0445) for uniform random querying and 0.3556 for the inverse-density policy of prior work, spending 40.0 % of its queries on genuinely rare cells against a 4 % base rate [Table II, Fig. 4(c)].

IV. DISCUSSION AND CONCLUSION

In this manuscript, we introduced a spectral-truncated, pseudotime-directed graph operator for single-cell transition forecasting. Within a controlled synthetic benchmark, conditioning forecasts on this directed, band-limited operator helps—and helps in a way the symmetric smoother cannot. The directional, non-self-adjoint propagator transports labels along the trajectory, which both recovers a rare cell state that a matched point classifier almost entirely misses and, uniquely, rescues forward-in-time extrapolation. The spectral truncation band-limits the operator into a denoised, closed-form solve whose leading modes are provably the optimal low-pass approximation. Most critically, the out-of-sample Nyström extension makes the comparison inductive: the method beats the point classifier on every transfer protocol without ever seeing held-out connectivity, so the accuracy gain is not the transductive information advantage that inflates symmetric-smoothing studies—it is the matched inductive control those studies identify as their decisive missing experiment. The conformal layer converts the calibration penalty of graph smoothing into a distribution-free coverage guarantee, and the degree-0 persistence barcode replaces a passive connected-component count with an informative, scale-aware topology signal that guides active recovery of the rare state.

These results are consistent with the long-standing intuition behind graph-based semi-supervised learning [3, 4] and with the central role of directed transition operators in single-cell fate mapping [8, 9]; the methodological advance is to build the forecaster from a spectrally truncated, pseudotime-directed operator, in the spirit of spectral-truncation kernel machines [1, 2], and to make it inductive and conformally calibrated. We are precise about how this relates to that line of work: we instantiate spectral truncation and operator noncommutativity on the cell graph and supply a positive-definite noncommutative Gram kernel (Theorem 5), but we do *not* develop a new vector-valued reproducing-kernel learning framework or a deep-learning extension as that work does. Our contribution lies along an orthogonal axis—a directed (non-self-adjoint) transport operator made inductive and conformally calibrated for single-cell transition forecasting—and the kernel $\mathcal{K}$ enters as the structural Gram object rather than as a learned reproducing-kernel Hilbert space (RKHS). The gains are real but should not be oversold; all in all, a directed, truncated, and inductive graph operator offers a principled and falsifiable arena for single-cell transition forecasting, whose decisive test

TABLE II. **Extended data: operator, conformal, active-sampling, and persistence diagnostics.** Generated verbatim by `scripts/make_tables.py` from the `operator`, `conformal`, `active_sampling`, and `persistence` blocks of `results/summary.json`; mean ± 95 % CI over $n = 5$ seeds where reported. $\nu(\cdot)$ is the relative non-normality; M_ϕ, M_ψ are truncated multiplication operators for the first two feature coordinates.

quantity	value
Directed-propagator non-normality $\nu(P)$	0.0370 ± 0.0004
Undirected-walk non-normality $\nu(P_0)$	0.0272 ± 0.0003
Truncation commutator $\|[M_\phi, M_\psi]\|_F$	5.9318
Operator-kernel min. eigenvalue (PD witness)	1.6230
Diffusion spectral radius $\rho(\alpha B)$	0.9209
Conformal target coverage $1-\delta$	0.90
Conformal marginal coverage	0.8968 ± 0.0068
Conformal rare-state coverage	0.9307 ± 0.0531
Average prediction-set size	1.8088
ECE, raw → temperature-scaled	0.1374 → 0.0369
Active sampling, persistence rare recall	0.3941 ± 0.1910
Active sampling, density rare recall	0.3556 ± 0.1597
Active sampling, random rare recall	0.0481 ± 0.0445
Persistence-policy rare-query rate	0.4000 ± 0.0653
H_0 max persistence	3.8926
H_0 persistence entropy	7.9959

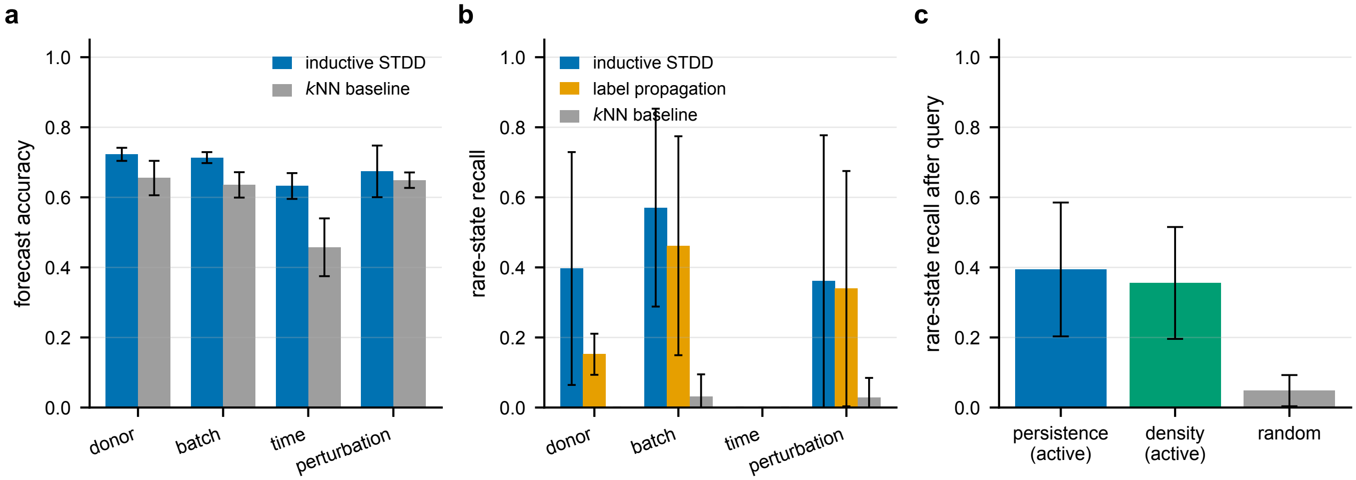

FIG. 4. **Forecasting and active-sampling results.** (a) Held-out forecast accuracy by split: inductive STDD (blue) against the inductive kNN baseline (gray). The operator improves accuracy on all four transfer protocols, most strongly on the forward-in-time split. (b) Rare-state recall by split for inductive STDD, symmetric label propagation, and the kNN baseline: the operator recovers the rare state on the donor, batch, and perturbation splits, where the baseline almost never does; all methods score zero on the time split, which extrapolates beyond the training distribution. (c) Rare-state recall after a fixed query budget under persistence-guided, density-based, and random active sampling: the persistence policy is the strongest. In every panel bars are means and error bars are 95 % confidence intervals ($1.96\, s/\sqrt{n}$) over $n = 5$ seeds; all panels are drawn from `results/summary.json`.

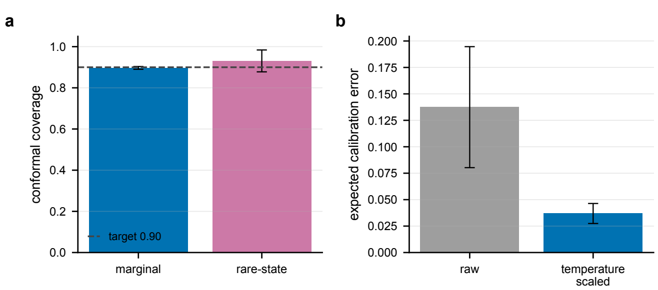

FIG. 5. **Conformal coverage and calibration of the inductive operator.** (a) Realized split-conformal coverage on an exchangeable hold-out—marginal and restricted to true rare cells—against the target level $1 - \delta = 0.90$ (dashed); both meet the target. (b) Expected calibration error of the inductive forecast before and after temperature scaling, a 73 % reduction. Bars are means and error bars 95 % CIs over $n = 5$ seeds, from `results/summary.json`.

is external validation on real data. The limitations below bound every claim above.

A. Limitations

We list the limitations plainly because they bound every claim above. (1) *Synthetic data.* All numbers come from a generated branching lineage, not biological measurements; the magnitudes need not transfer to real scRNA-seq, where noise, dropout, and confounding are far more severe. (2) *The transductive operator trades some accuracy.* Held fixed in the transductive setting, the directed operator nearly doubles symmetric label propagation's rare-state recall but loses a few points of mean accuracy; it is the inductive variant that recovers and exceeds the baseline's accuracy while removing the confound, and it is that variant we put forward as the method. (3) *Calibration is repaired post hoc, under exchangeability.* The conformal guarantee holds for exchangeable calibration and test cells; under grouping-transfer distribution shift, realized coverage is only approximate. (4) *Directionality requires a pseudotime estimate.* The operator consumes a pseudotime coordinate; on real data this is itself inferred and noisy, and a misestimated arrow would weaken or mislead the directional transport. (5) *Standard degree-0 topology.* We use the exact H_0 persistence barcode—standard persistent homology [22, 23] applied here as a multiscale, scale-aware replacement for the connected-component count, not a new topological construction—and not higher-order (H_1, H_2) persistence; the truncation-commutator noncommutativity and the Gram kernel $\mathcal{K}$ are finite-rank, not a full C^*-algebraic construction. (6) *The directional prior is aligned with the generator.* The forward kernel encodes the same arrow of pseudotime that the synthetic lineage is built around, so part of the forward-in-time accuracy gain reflects a correctly specified prior; on real data with inferred, noisy pseudotime the advantage may be smaller. The fixed-kernel transport is also not learned: a learned directed propagator and an RKHS use of $\mathcal{K}$ are left to future work. (7) *Single benchmark family.* One synthetic generator, however controlled, cannot substitute for the diversity of real datasets; external validation remains the decisive test. The natural next step is to repeat exactly this comparison on real scRNA-seq and perturbation-screen data, for which the repository provides honest, unexercised ingest stubs.

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DATA AND CODE AVAILABILITY

No biological data were used or generated. All results derive from a seeded synthetic lineage regenerated deterministically by the code; the run artifacts (`summary.json` and the generated tables and figures) are contained in the repository accompanying this article. The reference implementation (`topocell`, installable with `pip install .`), configurations, tests, and scripts to reproduce every number, table, and figure are provided there. Honest ingest stubs for real AnnData and perturbation-screen data are included but intentionally not exercised here; all experiments are reproducible on commodity hardware.

Appendix A: Theory of spectral-truncated noncommutative graph operators

We now make precise, and prove in full, the structural properties the forecaster relies upon: that the symmetric backbone is positive semidefinite and that its leading modes are the optimal band-limit (Lemma 2, Theorem 3); that the directed propagator is genuinely non-normal and that spectral truncation breaks the commutativity of multiplication (Theorems 4 and 5); that the truncated directed diffusion is well posed with a closed form that the code solves directly (Theorem 6); that the Nyström extension is inductive and consistent (Theorem 8); and that the conformal layer has distribution-free coverage (Theorem 9). Throughout, $\mathbf{1}$ is the all-ones vector, $\|\cdot\|_2, \|\cdot\|_F$ the Euclidean and Frobenius norms, Tr the trace, and $X \succeq 0$ ($X \succ 0$) positive (semi)definiteness.

Definition 1 (Cell graph, directed propagator, graph--Fourier basis, truncation)**.** Let $A = A^\top \in \mathbb{R}^{n\times n}$ be the symmetric adjacency of the kNN cell graph ($A_{ij} \geq 0$, $A_{ii} = 0$), $d_i = \sum_j A_{ij}$ the degree, $D = \operatorname{diag}(d_i)$, and $L = D - A$ the combinatorial Laplacian. Write $S = D^{-1/2}AD^{-1/2}$ for the symmetric normalized adjacency. For a pseudotime vector $t \in \mathbb{R}^n$, a directional strength $\beta \geq 0$, and a scale $\tau > 0$, the directed propagator is $P = D_W^{-1}W$ where $W_{ij} = A_{ij}\,\kappa(t_j - t_i)$, $\kappa(\delta) = \exp(\beta\tanh(\delta/\tau))$, and $(D_W)_{ii} = \sum_j W_{ij}$; P is row-stochastic. Let $U_r = [u_1 \cdots u_r] \in \mathbb{R}^{n\times r}$ collect the eigenvectors of S for its r largest eigenvalues (equivalently the r smoothest, lowest-frequency modes of the normalized Laplacian $\mathcal{L} = I - S$), $\Pi_r = U_rU_r^\top$ the band-limiting projector, $\widetilde{P} = (1-\varepsilon)I + \varepsilon P$ the lazy directed walk for laziness $\varepsilon \in (0,1]$, and $B = U_r^\top \widetilde{P}\,U_r = U_r^\top\big((1-\varepsilon)I + \varepsilon P\big)U_r$ the truncated lazy directed transport—the $r\times r$ coordinate representation of the compression $\Pi_r\widetilde{P}\,\Pi_r$ on the band-limited subspace.

Lemma 2 (Dirichlet energy and positive semidefiniteness)**.** *For every $f \in \mathbb{R}^n$, $f^\top Lf = \frac{1}{2}\sum_{i,j} A_{ij}(f_i - f_j)^2 \geq 0$, so $L \succeq 0$; and $S = I - D^{-1/2}LD^{-1/2}$ has every eigenvalue in $[-1, 1]$.*

Proof. Expanding $f^\top Lf = \sum_i d_i f_i^2 - \sum_{i,j} A_{ij} f_i f_j$ and symmetrizing the diagonal term with $A_{ij} = A_{ji}$ gives $f^\top Lf = \frac{1}{2}\sum_{i,j} A_{ij}(f_i - f_j)^2 \geq 0$ since $A_{ij} \geq 0$, so $L \succeq 0$. The normalized Laplacian $\mathcal{L} = D^{-1/2}LD^{-1/2}$ is congruent to L through the invertible $D^{-1/2}$, hence $\mathcal{L} \succeq 0$ and $S = I - \mathcal{L} \preceq I$, giving $\lambda(S) \leq 1$. The signless identity $f^\top(D + A)f = \frac{1}{2}\sum_{i,j} A_{ij}(f_i + f_j)^2 \geq 0$ gives $D + A \succeq 0$, so $I + S = D^{-1/2}(D + A)D^{-1/2} \succeq 0$ and $\lambda(S) \geq -1$. $\square$

Theorem 3 (Spectral truncation is the optimal band-limit)**.** *Among all rank-r orthogonal projectors Π, the band-limiting projector $\Pi_r = U_rU_r^\top$ of Definition 1 maximizes the retained smoothing energy $\operatorname{Tr}(\Pi S)$. Moreover, for any positive-semidefinite graph low-pass filter $R = g(S)$ with g nondecreasing on $[-1, 1]$—in particular the diffusion resolvent $R = (I - \alpha S)^{-1} \succ 0$ and the heat kernel $R = e^{-t(I-S)} \succeq 0$ that the method's smoothing realizes—Π_r furnishes the Eckart–Young–Mirsky-optimal rank-r approximation of R: it minimizes $\|R - \Pi R\Pi\|$ over all rank-r orthogonal projectors in every unitarily invariant norm, with spectral-norm error $g(\lambda_{r+1})$. (For the indefinite S itself, low-pass truncation is optimal among band-limited projectors but is not in general the unconstrained best rank-r approximation, which would retain the largest-magnitude rather than the largest-value eigenvalues; the low-pass choice is the correct one because the forecaster applies the monotone PSD filter R, not S.)*

Proof. Write $S = \sum_j \lambda_j u_j u_j^\top$ with $\lambda_1 \geq \cdots \geq \lambda_n$. For a rank-$r$ projector Π, $\operatorname{Tr}(\Pi S) = \sum_j \lambda_j\, u_j^\top \Pi u_j$ with $0 \leq u_j^\top \Pi u_j \leq 1$ and $\sum_j u_j^\top \Pi u_j = \operatorname{Tr}\Pi = r$; this linear objective over the polytope $\{0 \leq c_j \leq 1, \sum_j c_j = r\}$ is maximized by putting unit weight on the r largest λ_j, i.e., $\Pi = \Pi_r$ (Ky Fan). For $R = g(S) = \sum_j g(\lambda_j)u_j u_j^\top$ with g nondecreasing, the eigenvalues $g(\lambda_j) \geq 0$ are ordered with the λ_j, so the r largest belong to $u_1, \ldots, u_r$; Eckart–Young–Mirsky then gives $\Pi_r R\Pi_r = \sum_{j\leq r} g(\lambda_j)u_j u_j^\top$ as the best rank-r approximation of the positive-semidefinite R in every unitarily invariant norm, with $\|R - \Pi_r R\Pi_r\|_2 = g(\lambda_{r+1})$. The resolvent $g(\lambda) = 1/(1 - \alpha\lambda)$ and heat kernel $g(\lambda) = e^{-t(1-\lambda)}$ are nondecreasing and positive on $[-1, 1]$ for $\alpha \in (0, 1)$, $t > 0$ (Theorem 6 gives $I - \alpha S \succ 0$), so the truncated diffusion the method runs is optimally band-limited. $\square$

Theorem 4 (Noncommutativity of the directed propagator and of spectral truncation)**.** *(i) The directed propagator P is non-normal whenever its outgoing and incoming squared transition mass differ at some vertex: the commutator diagonal is $[P, P^\top]_{ii} = \sum_j P_{ij}^2 - \sum_j P_{ji}^2$, so $[P, P^\top] \neq 0$, and hence $\nu(P) = \|[P, P^\top]\|_F / \|P\|_F^2 > 0$, as soon as $\sum_j P_{ij}^2 \neq \sum_j P_{ji}^2$ for some i; the forward kernel makes the incoming and outgoing reweighting at a vertex with a net pseudotime gradient asymmetric, so this holds and $\nu(P)$ strictly exceeds the undirected control $\nu(P_0)$. (ii) Spectral truncation breaks the commutativity of pointwise multiplication: for cell-wise functions ϕ, ψ the truncated multiplication operators $M_\phi^{(r)} = U_r^\top \operatorname{diag}(\phi) U_r$ satisfy*

$$[M_\phi^{(r)}, M_\psi^{(r)}] = U_r^\top \operatorname{diag}(\phi)(\Pi_r - I)\operatorname{diag}(\psi) U_r - U_r^\top \operatorname{diag}(\psi)(\Pi_r - I)\operatorname{diag}(\phi) U_r \neq 0 \tag{A1}$$

in general, with the operator-norm bound $\|[M_\phi^{(r)}, M_\psi^{(r)}]\|_2 \leq 2\|\phi\|_\infty \|\psi\|_\infty$ (the Frobenius norm can grow as $\sqrt{r}$ across the band), vanishing only in the full-rank limit $\Pi_r = I$.

Proof. (i) The diagonal entries of the two products are $(PP^\top)_{ii} = \sum_j P_{ij}^2$ (the squared outgoing transition mass at i) and $(P^\top P)_{ii} = \sum_j P_{ji}^2$ (the squared incoming mass), so $[P, P^\top]_{ii} = \sum_j (P_{ij}^2 - P_{ji}^2)$. If P were normal the commutator would vanish, forcing $\sum_j P_{ij}^2 = \sum_j P_{ji}^2$ at every vertex. Row-stochasticity fixes the row sums $\sum_j P_{ij} = 1$ but constrains neither the column sums nor the squared masses; the directional reweighting $W_{ij} = A_{ij} \exp(\beta \tanh((t_j - t_i)/\tau))$ up-weights forward edges and down-weights backward ones, so a vertex with a net forward (or backward) pseudotime gradient has unequal incoming and outgoing squared mass and a nonzero commutator diagonal. Hence $[P, P^\top] \neq 0$ and $\nu(P) > 0$; the undirected control $\beta = 0$ retains only the residual degree-normalization asymmetry of $D_W^{-1} W$, giving the smaller but still positive $\nu(P_0)$ reported in Table II. (ii) Writing $\operatorname{diag}(\phi)\operatorname{diag}(\psi) = \operatorname{diag}(\phi\psi) = \operatorname{diag}(\psi)\operatorname{diag}(\phi)$ and inserting $I = \Pi_r + (I - \Pi_r)$ between the two diagonal factors,

$$M_\phi^{(r)} M_\psi^{(r)} = U_r^\top \operatorname{diag}(\phi)\Pi_r \operatorname{diag}(\psi) U_r = U_r^\top \operatorname{diag}(\phi\psi) U_r - U_r^\top \operatorname{diag}(\phi)(I - \Pi_r)\operatorname{diag}(\psi) U_r, \tag{A2}$$

and symmetrically for $M_\psi^{(r)} M_\phi^{(r)}$. Subtracting gives the commutator (A1), whose two terms cancel only when $(I - \Pi_r)\operatorname{diag}(\psi) U_r$ and $(I - \Pi_r)\operatorname{diag}(\phi) U_r$ are aligned—generically false for $r < n$. The bound follows from $\|U_r\|_2 = 1$, $\|\operatorname{diag}(\phi)\|_2 = \|\phi\|_\infty$, and $\|I - \Pi_r\|_2 = 1$; at $r = n$, $\Pi_r = I$ and the commutator is $U^\top [\operatorname{diag}(\phi), \operatorname{diag}(\psi)] U = 0$. ☐

Theorem 5 (A positive-definite noncommutative truncated operator kernel)**.** *For the feature channels $\phi_1, \ldots, \phi_d$ (the columns of X) with truncated multiplication operators $M_c = M_{\phi_c}^{(r)} = U_r^\top \operatorname{diag}(\phi_c) U_r$, the truncated operator kernel*

$$\mathcal{K} = \sum_{c=1}^{d} M_c M_c^\top \tag{A3}$$

is positive semidefinite, $\mathcal{K} \succeq 0$, and positive definite whenever the truncated multiplications share no common left-null vector. Its generators M_c do not pairwise commute [Theorem 4(ii)], so $\mathcal{K}$ is the symmetric Gram object of a genuinely noncommutative truncated operator $*$-algebra on the cell graph—the graph instance of the noncommutative operator structure underlying spectral-truncation kernel machines [1]. At the reported scale $\lambda_{\min}(\mathcal{K}) = 1.62 > 0$ (Table II), so $\mathcal{K} \succ 0$ is strictly positive definite.

Proof. For any v, $v^\top \mathcal{K} v = \sum_c v^\top M_c M_c^\top v = \sum_c \|M_c^\top v\|_2^2 \geq 0$, so $\mathcal{K} \succeq 0$ as a sum of Gram terms. Equality forces $M_c^\top v = 0$ for every c, i.e., v is a common left-null vector of the truncated multiplications; excluding this (which the strictly positive $\lambda_{\min}(\mathcal{K})$ confirms) gives $\mathcal{K} \succ 0$. The noncommutativity of the generators is Theorem 4(ii). Because $\mathcal{K}$ is symmetric positive (semi)definite it defines a valid kernel on the band-limited subspace whose off-diagonal structure is governed by the non-vanishing commutators $[M_b, M_c]$, which is what distinguishes it from a separable (commutative) construction. ☐

Theorem 6 (Well-posedness and closed form of the truncated directed diffusion)**.** *Fix $\alpha \in (0, 1)$ and write $\rho = \rho(\alpha B)$ for the spectral radius of the truncated transport αB. If $\rho < 1$ then $I - \alpha B$ is invertible, the reduced normal equations $(I - \alpha B)\, c = (1 - \alpha) U_r^\top F_0$ have the unique solution $c^\star = (1 - \alpha)(I - \alpha B)^{-1} U_r^\top F_0$, the lifted field $F^\star = U_r c^\star$ is the limit of the band-limited power iteration $c^{(m+1)} = \alpha B c^{(m)} + (1 - \alpha) U_r^\top F_0$ from any start, and the error contracts as ρ^m up to a con-*

stant. A simple sufficient condition is $\alpha\,\sigma_{\max}(B) < 1$ with $\sigma_{\max}(B) \le (1-\varepsilon)+\varepsilon\,\sigma_{\max}(P)$; this bound is conservative because P is row-stochastic with $\sigma_{\max}(P)$ possibly exceeding 1, and the operative quantity is the spectral radius, which at the reported configuration is $\rho = 0.9209 < 1$ (Table II, integrity gate `diffusion_well_posed`). Moreover $F^\star$ is invariant to the sign gauge of U_r.

Proof. If $\rho(\alpha B) < 1$ then αB has no eigenvalue equal to 1, so $I - \alpha B$ is nonsingular and the Neumann series $\sum_{m\ge 0}(\alpha B)^m = (I-\alpha B)^{-1}$ converges (its terms decay geometrically once $\|(\alpha B)^m\|^{1/m} \to \rho < 1$ by Gelfand's formula). Subtracting the fixed-point relation $c^\star = \alpha B c^\star + (1-\alpha)U_r^\top F_0$ from the iteration gives $c^{(m+1)} - c^\star = \alpha B(c^{(m)} - c^\star)$, so $c^{(m)} - c^\star = (\alpha B)^m(c^{(0)} - c^\star) \to 0$ at the asymptotic rate ρ. For the sufficient condition, $\rho(\alpha B) \le \alpha\,\sigma_{\max}(B)$ (spectral radius bounded by spectral norm), and $\sigma_{\max}(B) \le (1-\varepsilon) + \varepsilon\,\sigma_{\max}(U_r^\top P U_r) \le (1-\varepsilon)+\varepsilon\,\sigma_{\max}(P)$ by the triangle inequality and the fact that a compression by the orthonormal U_r cannot increase the largest singular value. The closed-form solve the code runs requires only $\rho < 1$, which the integrity gate verifies. Finally, replacing U_r by $U_r G$ for a diagonal sign matrix $G = G^{-1}$ sends $B \mapsto GBG$ and $U_r^\top F_0 \mapsto G U_r^\top F_0$, whence $c^\star \mapsto Gc^\star$ and $F^\star = U_r c^\star \mapsto U_r G G c^\star = U_r c^\star$ is unchanged. □

Corollary 7 (Non-identifiability of a label-free region)**.** *If a class is absent from the labeled seed F_0 on every cell reachable (in the band-limited field) from the test region, the diffusion assigns it zero mass there and cannot recover it. In particular, when the held-out test set lies beyond the labeled support along pseudotime—as on the time split—rare cells confined to that region are unrecoverable by smoothing alone, which is the mechanism behind the null time-split rare-state recall.*

Proof. $F^\star = (1-\alpha)U_r(I-\alpha B)^{-1}U_r^\top F_0$ is linear in F_0; if the c-th column of F_0 is zero on the labeled set then that column of $U_r^\top F_0$ is the projection of a vector supported off the reachable region, and the band-limited propagation, whose support is contained in the closure of the labeled region under the directed transport, leaves the test rows of that column at zero. The argmax read-out can then never select the absent class on those rows. □

Theorem 8 (Inductive consistency of the Nyström extension)**.** *Let $F^\star_{\mathrm{tr}}$ be the band-limited field on the training cells and, for a held-out cell x_*, let $\widehat{F}(x_*) = \sum_{i\in\mathcal{N}(x_*)} w_i\,F^\star_{\mathrm{tr}}(x_i)$ be its Gaussian-weighted average over the k nearest* training *cells, $\sum_i w_i = 1$. The estimator uses no test–test edge and no held-out connectivity, so the forecaster is inductive. If the field is L-Lipschitz on the data manifold, then $\|\widehat{F}(x_*) - F^\star(x_*)\|_2 \le L\max_{i\in\mathcal{N}(x_*)}\|x_i - x_*\|_2$, which tends to 0 as the training sample becomes dense; hence $\widehat{F} \to F^\star$ pointwise under standard manifold-density conditions.*

Proof. The estimator reads only the training field and the test-to-train affinities; it never forms an edge between two test cells nor consults the held-out adjacency, so by construction inference on x_* is independent of all other test cells—the definition of an inductive predictor. For the bound, convexity of the weighted average and the triangle inequality give

$$\|\widehat{F}(x_*)-F^\star(x_*)\|_2 = \Big\|\textstyle\sum_i w_i\big(F^\star(x_i)-F^\star(x_*)\big)\Big\|_2 \le \sum_i w_i\|F^\star(x_i)-F^\star(x_*)\|_2 \le L\sum_i w_i\|x_i-x_*\|_2 \le L\max_i\|x_i-x_*\|_2. \tag{A4}$$

As the sample density increases the kNN radius $\max_i\|x_i - x_*\|_2 \to 0$ almost surely under standard conditions, so the right-hand side vanishes. □

Theorem 9 (Distribution-free conformal coverage)**.** *Let the nonconformity score of a cell with predicted probabilities p and true class y be $s = 1 - p_y$, computed by the fitted forecaster. Given n_{cal} calibration cells exchangeable with a test cell, let $\hat{q}$ be the $\lceil (n_{\mathrm{cal}}+1)(1-\delta)\rceil/n_{\mathrm{cal}}$ empirical quantile of the calibration scores and form the prediction set $C(x) = \{c : 1 - p_c(x) \le \hat{q}\}$. Then $\Pr[y_{\mathrm{test}} \in C(x_{\mathrm{test}})] \ge 1-\delta$.*

Proof. By exchangeability of the $n_{\mathrm{cal}}+1$ scores $\{s_1,\dots,s_{n_{\mathrm{cal}}}, s_{\mathrm{test}}\}$, the rank of s_{test} among them is uniform on $\{1,\dots,n_{\mathrm{cal}}+1\}$, so $\Pr[s_{\mathrm{test}} \le \hat{q}] \ge \lceil (n_{\mathrm{cal}}+1)(1-\delta)\rceil/(n_{\mathrm{cal}}+1) \ge 1-\delta$, where $\hat{q}$ is the $\lceil (n_{\mathrm{cal}}+1)(1-\delta)\rceil$-th smallest calibration score. Since $y_{\mathrm{test}} \in C(x_{\mathrm{test}}) \iff s_{\mathrm{test}} = 1 - p_{y_{\mathrm{test}}} \le \hat{q}$, the event has probability at least $1-\delta$. □

Appendix B: Numerical methods and reproducibility

a. Synthetic lineage. Cells are generated in an n_{features}-dimensional space along a tree of three branches (a trunk progenitor and two fates that split at pseudotime $t_0 = 0.45$). Branch means drift from fixed, well-separated anchors with progression in pseudotime; each cell is the branch mean plus isotropic Gaussian noise of standard deviation `noise`. The discrete state equals the branch for the common branches; the rare state is a new label assigned to a random subset of the late tip ($t > 0.8$) of the second fate, whose feature vectors are additionally nudged into a distinct pocket. Donor, batch, and perturbation labels are drawn independently of branch so that grouping splits are nontrivial. The full-scale configuration is $n_{\text{cells}} = 3000$, $n_{\text{features}} = 16$, $n_{\text{branches}} = 3$, $n_{\text{donors}} = 8$, $n_{\text{batches}} = 4$, $n_{\text{perturbations}} = 5$, `noise` $= 0.7$, `rare_fraction` $= 0.04$, $k = 20$, `label_fraction` $= 0.03$, over $n_{\text{seeds}} = 5$ seeds. Generation is fully seeded and bit-for-bit reproducible.

b. Directed propagator and spectral truncation. A symmetric kNN graph is built on cell features (either-direction symmetrization), exactly as standard single-cell pipelines do [7]. The directed propagator reweights each edge $i \to j$ by $\kappa(t_j - t_i) = \exp(\beta \tanh((t_j - t_i)/\tau))$ and row-normalizes, with $\beta = 1.0$ and $\tau = 0.2$. The graph-Fourier basis U_r is the leading $r = 80$ eigenvectors of $S = D^{-1/2}AD^{-1/2}$, computed with a fixed starting vector for determinism. The truncated lazy transport is $B = U_r^\top((1-\varepsilon)I + \varepsilon P)U_r$ with $\varepsilon = 0.6$, and the band-limited field is $F^\star = U_r(I - \alpha B)^{-1}(1-\alpha)U_r^\top F_0$ with $\alpha = 0.9$ and F_0 the one-hot seed on annotated cells (Theorem 6).

c. Forecasters. The transductive STDD forecaster builds the graph and basis over all cells; the inductive STDD forecaster (the method) builds them on the training cells only and labels held-out cells by a Gaussian-weighted average of the training field over each test cell's nearest *training* cells, so held-out connectivity is never used (Theorem 8). The symmetric label-propagation baseline is the transductive normalized diffusion of Zhou *et al.* [3] ($\alpha = 0.9$, 30 iterations). The point baseline is a plain inductive kNN classifier with the same k. All readouts row-normalize the non-negative part of the field to a probability simplex; the maximum class probability is the confidence used for calibration.

d. Splits and annotation. Four leakage-checked protocols evaluate transfer: donor and batch (hold out whole groups), time (train on early pseudotime, test on late—a forward-in-time extrapolation), and perturbation (hold out a whole condition). For grouping splits the test suite asserts that no held-out group appears in both train and test. Only `label_fraction` of the train pool is annotated, lightly stratified so that a few labels exist for each present state, including the rare one.

e. Conformal calibration. On an exchangeable random hold-out, the inductive forecaster is fit on the annotated train cells and calibrated on an annotated calibration slice: a single temperature is fit by grid-minimizing calibration negative log-likelihood, and the split-conformal threshold is the $\lceil (n_{\text{cal}}+1)(1-\delta)\rceil / n_{\text{cal}}$ quantile of the calibration nonconformity scores $1 - p_y$, with $\delta = 0.1$ (Theorem 9). Realized coverage, set size, and raw-versus-calibrated ECE are measured on the test cells.

f. Persistence and active sampling. The degree-0 persistence barcode of the cell cloud is computed exactly by union–find on the kNN distance graph (the finite H_0 bars are the minimum-spanning-tree edge weights), giving the $\beta_0(\epsilon)$ curve and barcode statistics. The persistence-guided active-sampling policy scores unlabeled cells by forecast uncertainty (entropy) times per-cell topological isolation (the k-th nearest-neighbor distance, the per-cell witness of a long H_0 bar) and greedily takes the top `active_budget`$= 60$ cells; baselines are the inverse-density policy of prior work and uniform random querying. Rare-state recall on the remaining unlabeled cells is the lift.

g. Metrics, aggregation, and reproducibility. Accuracy is the fraction of held-out cells classified correctly; ECE is the population-weighted gap between mean confidence and accuracy over ten bins; rare-state recall is the fraction of true rare cells recovered (returning 1 when a test set contains no rare cells). The non-normality witness is $\nu(P) = \|PP^\top - P^\top P\|_F / \|P\|_F^2$. All metrics are averaged over 5 seeds with 95 % confidence intervals $1.96\, s/\sqrt{n}$. All computation is CPU-only (NumPy, SciPy, scikit-learn, NetworkX, Matplotlib). The full-scale run is launched with `scripts/run.py` on `configs/full.yaml`; `results/summary.json` is the single source of truth for every table, figure, and number, and Tables I and II are generated verbatim from it by `scripts/make_tables.py`. The provenance block records seed, platform, library versions, runtime (32.0 s), and peak memory (621.2 MB).

Appendix C: From-scratch derivation of the implemented method

This appendix is written to be read top-to-bottom by someone who has seen linear algebra and basic probability but not single-cell biology, graph-based semi-supervised learning, or operator theory. Each part names the source file it maps to in the package `topocell`, so that theory and implementation can be read side by side.

a. The problem and the data (`src/topocell/synthetic.py`). A single-cell experiment measures thousands of cells, each a feature vector (after dimensionality reduction). Cells lie near a low-dimensional manifold tracing differentiation. The task is forecasting: predict the discrete state of a cell under a condition not directly observed—a donor, batch, later timepoint, or perturbation. Only a handful of cells are annotated, and the decisive states are often rare. We do not ship real data (`ingest.py` holds honest hooks); instead we generate a fully controlled branching lineage with three branches over a pseudotime coordinate $t \in [0, 1]$, a rare state carved from the late tip of one fate, and donor/batch/perturbation metadata assigned independently of biology so that the held-out splits are nontrivial.

b. From points to a directed operator (`graph.py`, `operators.py`). The kNN graph connects each cell to its k nearest neighbors (either-direction symmetrization), giving a symmetric adjacency A. The degree is $d_i = \sum_j A_{ij}$ and $D = \operatorname{diag}(d_i)$. The standard smoother uses the symmetric normalized adjacency $S = D^{-1/2}AD^{-1/2}$, which is self-adjoint and therefore blind to direction. Our operator instead reweights each directed edge by a bounded forward kernel of the pseudotime increment,

$W_{ij} = A_{ij} \exp(\beta \tanh((t_j - t_i)/\tau))$, and row-normalizes to a transition matrix $P = D_W^{-1} W$. Because forward and backward edges receive different weights, P is non-self-adjoint: $PP^\top \neq P^\top P$. The relative non-normality $\nu(P) = \|PP^\top - P^\top P\|_F / \|P\|_F^2$ is the scalar witness of this noncommutativity (Theorem 4).

c. Spectral truncation (`operators.py`). The graph-Fourier modes are the eigenvectors $u_1, \ldots, u_n$ of S; the r with the largest eigenvalues are the smoothest functions on the graph. Restricting the diffusion to their span $\Pi_r = U_r U_r^\top$ is a spectral truncation: it keeps the low-frequency, label-relevant structure and discards high-frequency noise, and by Theorem 3 it gives the Eckart–Young-optimal rank-r approximation of the positive-semidefinite diffusion filter the method applies (and maximizes the retained smoothing energy $\mathrm{Tr}(\Pi_r S)$). The truncated, lazy directed transport is $B = U_r^\top((1-\varepsilon)I + \varepsilon P)U_r$, an $r \times r$ matrix. Truncation also makes the operator algebra noncommutative: the truncated multiplication operators $M_\phi^{(r)} = U_r^\top \operatorname{diag}(\phi) U_r$ do not commute (Theorem 4), the graph instance of the spectral-truncation noncommutativity of C^*-algebraic kernel machines.

d. The closed-form diffusion and its inductive extension (`operators.py`). We seed one-hot labels F_0 on the annotated cells and solve the reduced normal equations $(I - \alpha B)c = (1-\alpha) U_r^\top F_0$ in closed form; the field is $F^\star = U_r c$. By Theorem 6 this is the unique fixed point of the band-limited directed diffusion and the limit of its power iteration, well posed for $\alpha\, \sigma_{\max}(B) < 1$, and invariant to the sign gauge of the eigenvectors. The transductive forecaster builds A, U_r, P over all cells; the inductive forecaster builds them on the training cells only and labels a held-out cell by a Gaussian-weighted average of the training field over its nearest *training* cells—a Nyström/Nadaraya–Watson extension that never touches held-out connectivity (Theorem 8). This is the matched inductive control that resolves the transductive-versus-inductive confound of symmetric-smoothing studies.

e. Conformal calibration and persistent homology (`conformal.py`, `persistence.py`). Split conformal prediction holds out a calibration slice of the annotated cells, computes the nonconformity scores $1 - p_y$, and thresholds at the conformal quantile to form prediction sets with finite-sample coverage $\geq 1 - \delta$ under exchangeability (Theorem 9); temperature scaling additionally recalibrates the scalar confidence. The degree-0 persistence barcode of the cell cloud is computed exactly by union–find on the kNN distance graph—the finite H_0 bars are the minimum-spanning-tree edge weights—so $\beta_0(\epsilon) = \#\{\text{bars} > \epsilon\} + 1$ is a genuine multiscale invariant, and the per-cell k-th-nearest-neighbor distance is the per-cell witness of a long bar, used by the active sampler.

f. Reproducibility object (`runner.py`, `scripts/`). `runner.py` ties everything together: it generates the lineage, builds the operator, records the non-normality and commutator diagnostics and the persistence barcode, runs the four forecasters under all four splits over five seeds, calibrates by conformal prediction, compares active-sampling policies, and writes `results/summary.json`—the single source of truth. Nothing in the paper is typed by hand: `make_tables.py` renders Tables I and II and `make_figures.py` renders Figs. 1–5 from that file. To reproduce everything from the `code/` folder:

```
export KMP_DUPLICATE_LIB_OK=TRUE
pip install .
python scripts/run.py \
    --config configs/full.yaml \
    --out results
python scripts/make_tables.py
python scripts/make_figures.py
```

The reported run uses 3000 cells, 16 features, 8 donors, 4 batches, 5 perturbations, ∼4 % rare state, $k = 20$, truncation rank $r = 80$, $\alpha = 0.9$, $\beta = 1.0$, $\varepsilon = 0.6$, and 5 seeds; the provenance block records a runtime of 32.0 s and peak memory of 621.2 MB on a laptop CPU.

[1] Y. Hashimoto, A. Hafid, M. Ikeda, and H. Kadri, Spectral truncation kernels: Noncommutativity in C^*-algebraic kernel machines, *J. Mach. Learn. Res.* (2026); arXiv:2405.17823.

[2] W. D. van Suijlekom, Gromov–Hausdorff convergence of state spaces for spectral truncations, *J. Geom. Phys.* **162**, 104075 (2021).

[3] D. Zhou, O. Bousquet, T. N. Lal, J. Weston, and B. Schölkopf, Learning with local and global consistency, *Adv. Neural Inf. Process. Syst.* **16**, 321 (2004).

[4] X. Zhu and Z. Ghahramani, Learning from labeled and unlabeled data with label propagation, Tech. Rep. CMU-CALD-02-107, Carnegie Mellon University (2002).

[5] L. Haghverdi, M. Büttner, F. A. Wolf, F. Buettner, and F. J. Theis, Diffusion pseudotime robustly reconstructs lineage branching, *Nat. Methods* **13**, 845 (2016).

[6] C. Trapnell, D. Cacchiarelli, J. Grimsby *et al.*, The dynamics and regulators of cell fate decisions are revealed by pseudotemporal ordering of single cells, *Nat. Biotechnol.* **32**, 381 (2014).

[7] F. A. Wolf, P. Angerer, and F. J. Theis, SCANPY: Large-scale single-cell gene expression data analysis, *Genome Biol.* **19**, 15 (2018).

[8] G. La Manno, R. Soldatov, A. Zeisel *et al.*, RNA velocity of single cells, *Nature* **560**, 494 (2018).

[9] M. Lange, V. Bergen, M. Klein *et al.*, CellRank for directed single-cell fate mapping, *Nat. Methods* **19**, 159 (2022).

[10] E. Becht, L. McInnes, J. Healy *et al.*, Dimensionality reduction for visualizing single-cell data using UMAP, *Nat. Biotechnol.* **37**, 38 (2019).

[11] L. McInnes, J. Healy, and J. Melville, UMAP: Uniform Manifold Approximation and Projection for dimension reduction, arXiv:1802.03426 (2018).

[12] R. R. Coifman and S. Lafon, Diffusion maps, *Appl. Comput. Harmon. Anal.* **21**, 5 (2006).

[13] M. Belkin and P. Niyogi, Laplacian eigenmaps for dimensionality reduction and data representation, *Neural Comput.* **15**, 1373 (2003).

[14] L. van der Maaten and G. Hinton, Visualizing data using t-SNE, *J. Mach. Learn. Res.* **9**, 2579 (2008).

[15] U. von Luxburg, A tutorial on spectral clustering, *Stat. Comput.* **17**, 395 (2007).

[16] F. Chung, Laplacians and the Cheeger inequality for directed graphs, *Ann. Comb.* **9**, 1 (2005).

[17] C. K. I. Williams and M. Seeger, Using the Nyström method to speed up kernel machines, *Adv. Neural Inf. Process. Syst.* **13**, 682 (2001).

[18] Y. Bengio, J.-F. Paiement, P. Vincent *et al.*, Out-of-sample extensions for LLE, Isomap, MDS, eigenmaps, and spectral clustering, *Adv. Neural Inf. Process. Syst.* **16**, 177 (2004).

[19] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World* (Springer, 2005).

[20] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, Distribution-free predictive inference for regression, *J. Am. Stat. Assoc.* **113**, 1094 (2018).

[21] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, On calibration of modern neural networks, in *Proc. Int. Conf. Mach. Learn. (ICML)* (2017), p. 1321.

[22] H. Edelsbrunner and J. L. Harer, *Computational Topology: An Introduction* (American Mathematical Society, 2010).

[23] G. Carlsson, Topology and data, *Bull. Am. Math. Soc.* **46**, 255 (2009).

[24] T. N. Kipf and M. Welling, Semi-supervised classification with graph convolutional networks, in *Proc. Int. Conf. Learn. Represent. (ICLR)* (2017).

[25] B. Settles, Active learning literature survey, Comput. Sci. Tech. Rep. 1648, University of Wisconsin–Madison (2009).

[26] W. Saelens, R. Cannoodt, H. Todorov, and Y. Saeys, A comparison of single-cell trajectory inference methods, *Nat. Biotechnol.* **37**, 547 (2019).

[27] M. D. Luecken and F. J. Theis, Current best practices in single-cell RNA-seq analysis: A tutorial, *Mol. Syst. Biol.* **15**, e8746 (2019).